

Vojtěch Špaček

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Senior C++ / Python engineer — safety-critical embedded, HIL verification & agentic-AI (MCP) tooling | Brno · independent contractor (IČO)

Profile

- Sole owner of the **HIL verification suite and automated bench** for a safety-certified gas furnace ignition controller (UL/CSA, ANSI Z21.20, NFPA 54).
- Ten years across **aerospace** (Honeywell), **scientific instruments** (Thermo Fisher) and **network security** (Kernun firewall, Trusted Network Solutions).
- I bring **agentic-AI tooling (Model Context Protocol)** into safety-critical engineering workflows.

Availability

- **Independent contractor on a Czech IČO** — Brno on-site, hybrid or fully remote; available for long-term engagements and day-rate work.
- **EU citizen (Czech Republic)**; Czech and Slovak native, English C1, German basic.

Core Skills

- **C++** (*Senior*, C++17/20) · **Python** (*Senior*) · **C#** (*Medior*) · **JavaScript** (*Medior*) · **Rust**, **Qt/QML** (*working knowledge*)
- Embedded & RTOS, ARM, state-machine design, HAL & IPC, Flatbuffers, Conan, gRPC
- Hardware-in-the-loop verification, [pyvisa](#)/[USBTMC](#)/[SCPI](#), [CAN](#), [Modbus](#), requirement traceability
- Linux, [Netlink](#), [nftables](#), TLS inspection, Kerberos/LDAP/OCSP, PostgreSQL
- Model Context Protocol, agentic workflows, LLM provider APIs, CI/CD for AI-assisted delivery

Experience

Resideo (Ademco) - CoolPeople Contractor

[Test Automation Engineer \(Python, C#\)](#) (01/2026 – 09/2026): Hardware-in-the-loop verification of a safety-certified two-stage gas furnace ignition controller ([UL/CSA](#), [ANSI Z21.20](#), [NFPA 54](#)), as sole owner of the verification test suite and of the automated bench that drives it.

- Designed an [Arrange/Act/Assert](#) validation pipeline in [Python](#) with eight independent gates covering preconditions, stimulus timing, state sequence, alarms, signal history, cross-layer consistency, timer accuracy and requirement traceability.
- Built a data-driven generator expanding a [YAML](#) catalogue of 270+ state transitions, 38 states and 33 events into roughly 290 executable tests, removing hand-written duplication.
- Engineered the bench device layer so each instrument is owned by a single thread against a shared monotonic clock, covering [pyvisa](#)/[USBTMC](#)/[SCPI](#) programmable AC source, [CAN](#) serial-bus monitoring and [Modbus](#) relay and probe I/O.
- Implemented a correlated [JSONL](#) timeline placing every instrument command, signal edge and bus message on one axis, enabling root-cause analysis across three independent devices.

Thermo Fisher Scientific - Ixperta Contractor

Software Developer (C++): Electron Microscopy software development for high-resolution detector hardware operation and image acquisition.

- Implemented server-side software in **C++**, including **hardware abstraction layers (HAL)** and **inter-process communication (IPC)** using **COM** and **MIDL** on Windows.
- Developed embedded software for **FPGA** hardware, utilizing network communication protocols (**gRPC**) in C++, Python, and Linux environments.

Honeywell

- **Project**: Onboard Maintenance System that monitors the condition of aircraft and diagnoses issues during flight.
- Architected and implemented **DiagTest framework**, a sophisticated **state machine-based diagnostic test engine** in **C++** managing 15+ operational states for aircraft member system testing during flight and maintenance operations.

Trusted Network Solutions

- **Implemented a forward proxy** using **C++** on **Linux**, featuring **HTTP/HTTPS proxy with TLS/SSL inspection**, **OCSP integration** for certificate validation, **Kerberos authentication**, and **LDAP integration** for user/group lookup.
- **Developed and maintained ipmon**, a sophisticated network monitoring solution demonstrating deep expertise in:

IBM — Automation Developer / IT Systems Specialist

- Automation application replacing manual ticketing and server-administration workload (**Bash**, **PowerShell**, **JavaScript**); monitored Unix/Windows production servers.

Education & Training

- **Masaryk University, Faculty of Informatics** — Computer Science coursework, *no degree awarded*.
- **Coursera / DeepLearning.AI** — Deep Learning Specialization (Andrew Ng), prompt engineering, building LLM agents. Self-directed: design patterns, Hack The Box.

Selected Open Source

- **MCP-Prompts** (97 ★, 21 forks) — Model Context Protocol server for prompt templates and LLM interactions. On npm, Docker Hub and the MCP directories; Rust port on crates.io.
- **rust-network-mgr** — Rust daemon driving **nftables** rules via **Netlink**; lock-free, epoll-based. On crates.io.
- **ipmon** — C++20 Netlink monitoring daemon; RAIL, thread-safe atomics, Debian-packaged with systemd hardening.